

GE23131-Programming Using C-2024

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Duration	45 days 21 hours

Question 1

Correct

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Alice and Bob are playing a game called "Stone Game". Stone game is a two-player game. Let N be the total number of stones. In each turn, a player can remove either one stone or four stones. The player who picks the last stone, wins. They follow the "Ladies First" norm. Hence Alice is always the one to make the first move. Your task is to find out whether Alice can win, if both play the game optimally.

Input Format

First line starts with T , which is the number of test cases. Each test case will contain N number of stones.

Output Format

Print "Yes" in the case Alice wins, else print "No".

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Print "Yes" in the case Alice wins, else print "No".

Constraints

$$1 \leq T \leq 1000$$
$$1 \leq N \leq 10000$$

Sample Input and Output

Input

3

1

6

7

Output

Yes

Yes

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Yes

No

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main(){
3     int T,N,x;
4     scanf("%d", &T);
5     for (int i=1;i<=T;i++){
6         scanf("%d", &N);
7         x=N/4;
8         if((N % 2 == 0 && x % 2 != 0) || N == 1 || N == 3){
9             printf("Yes\n");
10        }
11        else{
12            printf("No\n");
13        }
14    }
15    return 0;
16 }
```

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```
14 | }  
15 |     return 0;  
16 | }
```

	Input	Expected	Got	
✓	3	Yes	Yes	✓
	1	Yes	Yes	
	6	No	No	
	7			

Passed all tests! ✓

Question 2

Correct

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You are designing a poster which prints out numbers with a unique style applied to each of them. The styling is based on the number of closed paths or holes present in a given number.

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The number of holes that each of the digits from 0 to 9 have are equal to the number of closed paths in the digit. Their

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1, 4, 6, 8, and 9 = 1 hole.

0, 4, 6, and 9 = 1 hole.

8 = 2 holes.

Given a number, you must determine the sum of the number of holes for all of its digits. For example, the number 819 has 3 holes.

Complete the program, it must must return an integer denoting the total number of holes in num.

Constraints

$1 \leq \text{num} \leq 109$

Input Format For Custom Testing

There is one line of text containing a single integer num, the value to process.

Sample Input

630

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Sample input

630

Sample Output

2

Explanation

Add the holes count for each digit, 6, 3 and 0. Return $1 + 0 + 1 = 2$.

Sample Case 1

Sample Input

1288

Sample Output

4

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Add the holes count for each digit, 1, 2, 8, 8. Return $0 + 0 + 2 + 2 = 4$.

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main(){
3     int num,totalholes=0;
4     int holes[10]={1,0,0,0,1,0,1,0,2,1};
5     scanf("%d",&num);
6     while(num>0){
7         int digit = num%10;
8         totalholes+=holes[digit];
9         num=num/10;
10    }
11    printf("%d\n",totalholes);
12    return 0;
13
14 }
```

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Input	Expected	Got

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```
6 * while(num>0){
7     int digit = num%10;
8     totalholes+=holes[digit];
9     num=num/10;
10 }
11 printf("%d\n",totalholes);
12 return 0;
13
14 }
```

	Input	Expected	Got	
✓	630	2	2	✓
✓	1288	4	4	✓

Passed all tests! ✓

Question 3

The problem solvers have found a new Island for coding and named it as Philaland. These smart people were given a task

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Passed all tests! ✓

Question 3

Correct

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Flag question

The problem solvers have found a new Island for coding and named it as Philaland. These smart people were given a task to make a purchase of items at the Island easier by distributing various coins with different values. Manish has come up with a solution that if we make coins category starting from \$1 till the maximum price of the item present on Island, then we can purchase any item easily. He added the following example to prove his point.

Let's suppose the maximum price of an item is 5\$ then we can make coins of {\$1, \$2, \$3, \$4, \$5} to purchase any item ranging from \$1 till \$5.

Now Manisha, being a keen observer suggested that we could actually minimize the number of coins required and gave following distribution {\$1, \$2, \$3}. According to him any item can be purchased one time ranging from \$1 to \$5. Everyone was impressed with both of them. Your task is to help Manisha come up with a minimum number of denominations for any arbitrary max price in Philaland.

Input Format

Contains an integer N denoting the maximum price of the item present on Philaland.

Output Format

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Contains an integer N denoting the maximum price of the item present on Philaland.

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Output Format

Print a single line denoting the minimum number of denominations of coins required.

Denoting an integer N denoting the maximum price of the item present on Philaland.

Constraints

Output Format

$1 \leq N \leq 5000$

Refer the sample output for formatting

Output Format

Sample input 1:

Refer the sample output for formatting

Sample input 1:

4
10

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Sample Input 2:

5

Sample Output 2:

3

Explanation:

For test case 1, $N=10$.

According to Manish $\{\$1, \$2, \$3, \dots \$10\}$ must be distributed.

But as per Manisha only $\{\$1, \$2, \$3, \$4\}$ coins are enough to purchase any item ranging from $\$1$ to $\$10$. Hence minimum is 4. Likewise denominations could also be $\{\$1, \$2, \$3, \$5\}$. Hence answer is still 4.

For test case 2, $N=5$.

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But as per Manisha only { \$1, \$2, \$3 } coins are enough to purchase any item ranging from \$1 to \$5. Hence minimum is 3. Likewise, denominations could also be { \$1, \$2, \$4 }. Hence answer is still 3.

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main(){
3     int n,c;
4     scanf("%d",&n);
5     int sum = 1;
6     while(sum<=n){
7         sum*=2;
8         c++;
9     }
10    printf("%d",c);
11 }
```

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	Input	Expected	Got	
✓	10	4	4	✓
✓	5	3	3	✓
✓	20	5	5	✓
✓	500	9	9	✓
✓	1000	10	10	✓

Passed all tests! ✓

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1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126 127 128 129 130 131 132 133 134 135 136 137 138 139 140 141 142 143 144 145 146 147 148 149 150 151 152 153 154 155 156 157 158 159 160 161 162 163 164 165 166 167 168 169 170 171 172 173 174 175 176 177 178 179 180 181 182 183 184 185 186 187 188 189 190 191 192 193 194 195 196 197 198 199 200 201 202 203 204 205 206 207 208 209 210 211 212 213 214 215 216 217 218 219 220 221 222 223 224 225 226 227 228 229 230 231 232 233 234 235 236 237 238 239 240 241 242 243 244 245 246 247 248 249 250 251 252 253 254 255 256 257 258 259 260 261 262 263 264 265 266 267 268 269 270 271 272 273 274 275 276 277 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2622 2623 2624 2625 2626 2627

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The value of the numbers can be from 00000000 to 99999999

Output Format:

The value of the numbers can be from 00000000 to 99999999

The count of numbers where the numbers are odd numbers.

Output Format:

Example Input / Output 1:

Input:

5 10 15 20 25 30 35 40 45 50
5 10 15 20 25 30 35 40 45 50

Output:

5
5

Explanation:

The numbers meeting the criteria are 5, 15, 25, 35, 45.

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The numbers meeting the criteria are 5, 15, 25, 35, 45.

Answer: (penalty regime: 0 %)

```
1 #include <stdio.h>
2 int main(){
3     int num,count=0,num_count=0;
4     while((scanf("%d",&num)==1)){
5         if(num%2!=0){
6             count++;
7         }
8         num_count++;
9     }
10    printf("%d",count);
11    return 0;
12 }
```

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	Input	Expected	Got	
✓	5 10 15 20 25 30 35 40 45 50	5	5	✓

Passed all tests! ✓

Question 2

Correct

Marked out of 5.00

Flag question

Given a number N, return true if and only if it is a *confusing number*, which satisfies the following condition:

We can rotate digits by 180 degrees to form new digits. When 0, 1, 6, 8, 9 are rotated 180 degrees, they become 0, 1, 9, 8, 6 respectively. When 2, 3, 4, 5 and 7 are rotated 180 degrees, they become invalid. A *confusing number* is a number that when rotated 180 degrees becomes a **different** number with each digit valid.

Example 1:

6 -> 9

Input: 6

Output: true

Explanation:

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Example 1:

6 -> 9

Output: true

Input: 6

Explanation:

Output: true

Explanation:

We get 9

after rotating 6, 9 is a valid number and 9!=6.

We get 9

Explanation:

Example 2:

89 -> 68

Input: 89

Output: false

Output: true

Explanation:

We get 68 after rotating 89, 68 is a valid number but the value remains the same, thus 89 is not a confusing number.

We get 68 after rotating 89, 68 is a valid number and 68!=89.

Example 3:

11 -> 11

Input: 11

Output: false

Explanation:

We get 11 after rotating 11, 11 is a valid number but the value remains the same, thus 11 is not a confusing number.

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2. After the rotation we can ignore leading zeros, for example if after rotation we have 0008 then this number is considered as just 8.

Answer: (penalty regime: 0 %)

```
1 #include <stdio.h>
2 int main(){
3     int a,ch;
4     scanf("%d",&a);
5     while(a!=0){
6         int b=a%10;
7         a=a/10;
8         switch(b){
9             case 0:
10                case 6:
11                case 8:
12                case 9:
13                    ch=0;
14                    break;
15                default:
16                    ch=1;
17            }
18        }
19        if(ch==1){
20            printf("false");
21        }
22        else{
23            printf("true");
24        }
25        return 0;
26 }
```

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	Input	Expected	Got	
✓	6	true	true	✓
✓	89	true	true	✓
✓	25	false	false	✓

Passed all tests! ✓

Question 3

Correct

Marked out of
7.00

Flag question

A nutritionist is labeling all the best power foods in the market. Every food item arranged in a single line, will have a value beginning from 1 and increasing by 1 for each, until all items have a value associated with them. An item's value is the same as the number of macronutrients it has. For example, food item with value 1 has 1 macronutrient, food item with value 2 has 2 macronutrients, and incrementing in this fashion.

The nutritionist has to recommend the best combination to patients, i.e. maximum total of macronutrients. However, the nutritionist must avoid prescribing a particular sum of macronutrients (an 'unhealthy' number), and this sum is known. The nutritionist chooses food items in the increasing order of their value. Compute the highest total of macronutrients that can be prescribed to a patient, without the sum matching the given 'unhealthy' number.

Here's an illustration:

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13/Jan/2025

Here's an illustration:

Given 4 food items (hence value: 1,2,3 and 4), and the unhealthy sum being 6 macronutrients, on choosing items 1, 2, 3 -> the sum is 6, which matches the 'unhealthy' sum. Hence, one of the three needs to be skipped. Thus, the best combination is from among:

- $2 + 3 + 4 = 9$
- $1 + 3 + 4 = 8$
- $1 + 2 + 4 = 7$

Since $2 + 3 + 4 = 9$, allows for maximum number of macronutrients, 9 is the right answer.

Complete the code in the editor below. It must return an integer that represents the maximum total of macronutrients, modulo 1000000007 ($10^9 + 7$).

It has the following:

n : an integer that denotes the number of food items

k : an integer that denotes the unhealthy number

Constraints

- $1 \leq n \leq 2 \times 10^9$
- $1 \leq k \leq 4 \times 10^{15}$

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The first line contains an integer, n , that denotes the number of food items.

The second line contains an integer, k , that denotes the unhealthy number.

Sample Input 0

2

2

Sample Output 0

3

Explanation 0

The following sequence of $n = 2$ food items:

1. Item 1 has 1 macronutrients.
2. $1 + 2 = 3$; observe that this is the max total, and having avoided having exactly $k = 2$ macronutrients.

Sample Input 1

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Explanation 1

1. Cannot use item 1 because $k = 1$ and $sum \equiv k$ has to be avoided at any time.
2. Hence, max total is achieved by $sum = 0 + 2 = 2$.

Sample Case 2

Sample Input For Custom Testing**Sample Input 2**

3

3

Sample Output 2

5

Explanation 2

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explanation :

$2 + 3 = 5$, is the best case for maximum nutrients.

Answer: (penalty regime: 0 %)

```
1 #include <stdio.h>
2 int main(){
3     long int n,k,sum;
4     scanf("%ld %ld",&n,&k);
5     sum=0;
6     for(int i=1;i<=n;i++){
7         sum=sum+i;
8         if(sum==k){
9             sum=sum-1;
10        }
11    }
12    printf("%ld",sum%1000000007);
13    return 0;
14 }
```

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```
13 | return 0;  
14 | }
```

	Input	Expected	Got	
✓	2 2	3	3	✓
✓	2 1	2	2	✓
✓	3 3	5	5	✓

Passed all tests! ✓

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